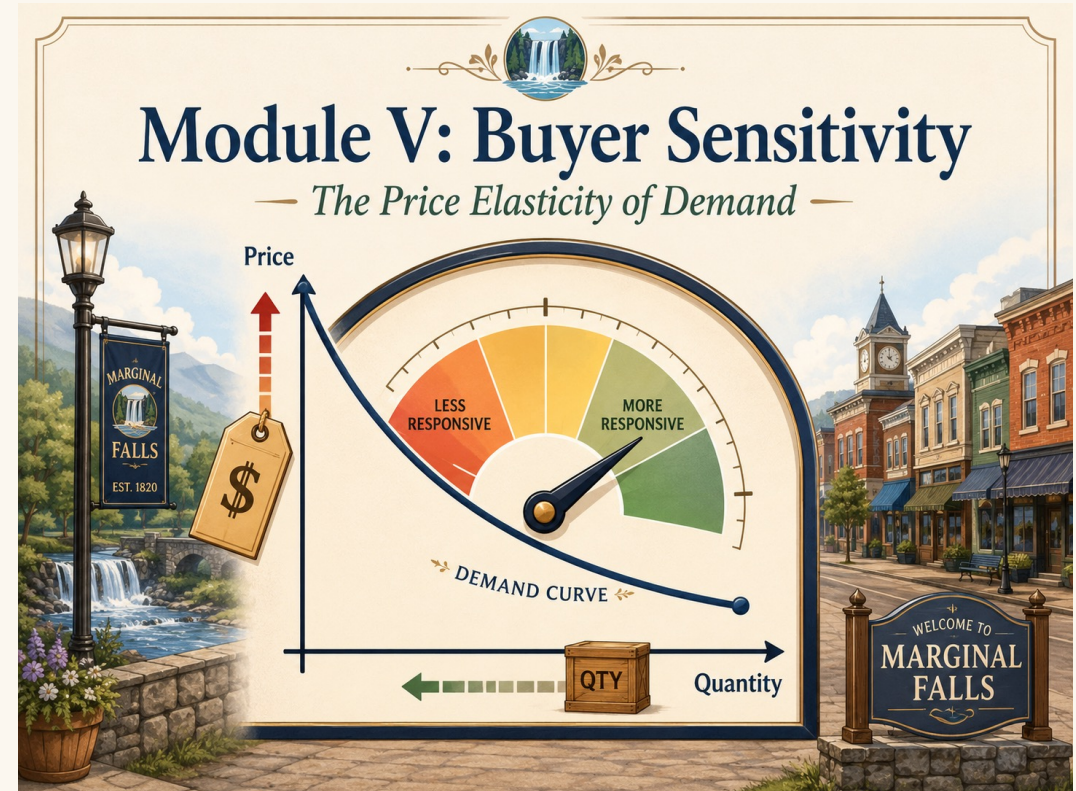


Act II: Casting the Buyers

Motivations and Choices of the Citizens

Module V: Buyer Sensitivity

Price Elasticity of Demand



Previously in Marginal Falls

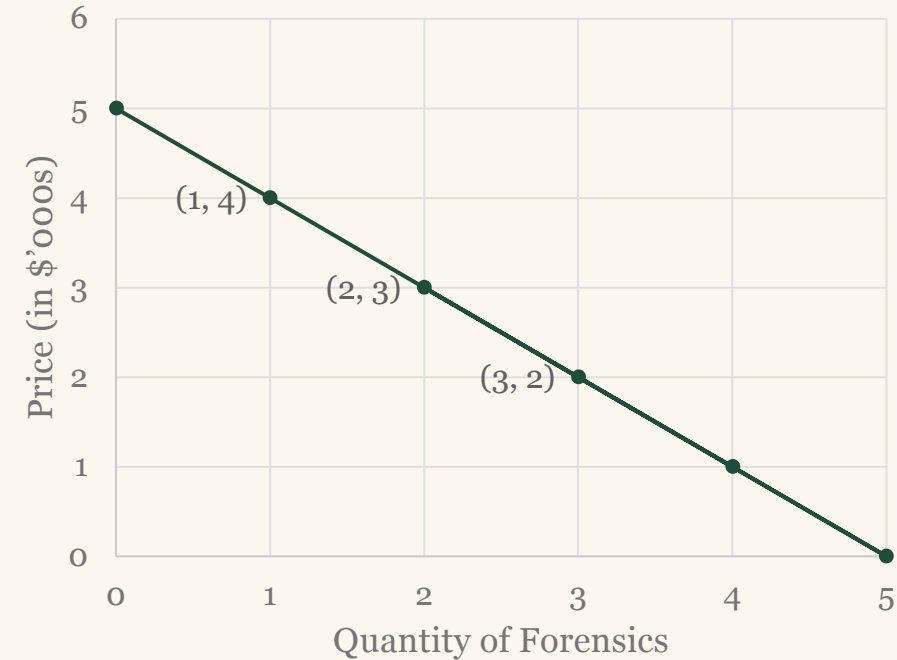
- ◆ **Utility** tells us what buyers want.
- ◆ **Budget constraints** tell us what they can afford.
- ◆ **Demand** tells us quantity demanded at each price.
- ◆ **Market Demand** is the aggregate demand at each price.

Next: Measure the responsiveness of demand.

Back to the Demand for Forensics

Price $\uparrow \rightarrow$ quantity demanded $\downarrow$

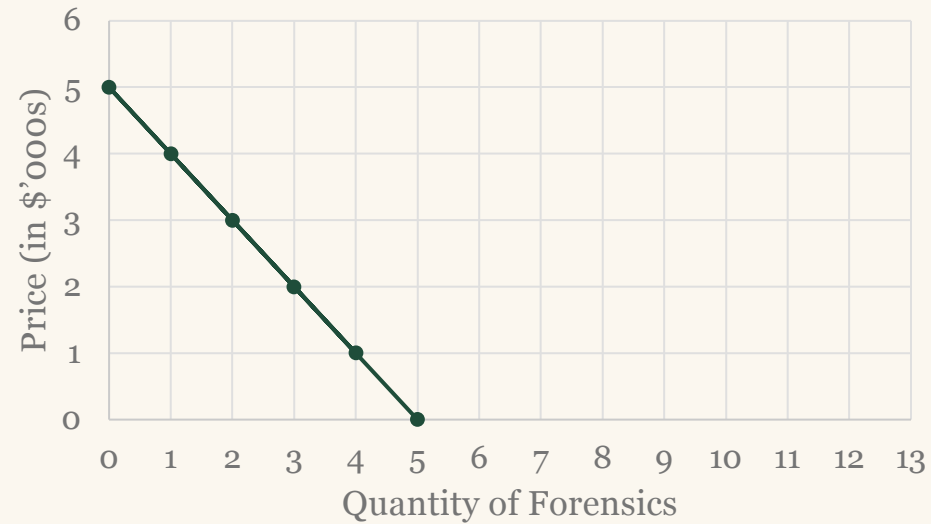
Price of Forensics	Forensics
\$1,000	4
\$2,000	3
\$3,000	2



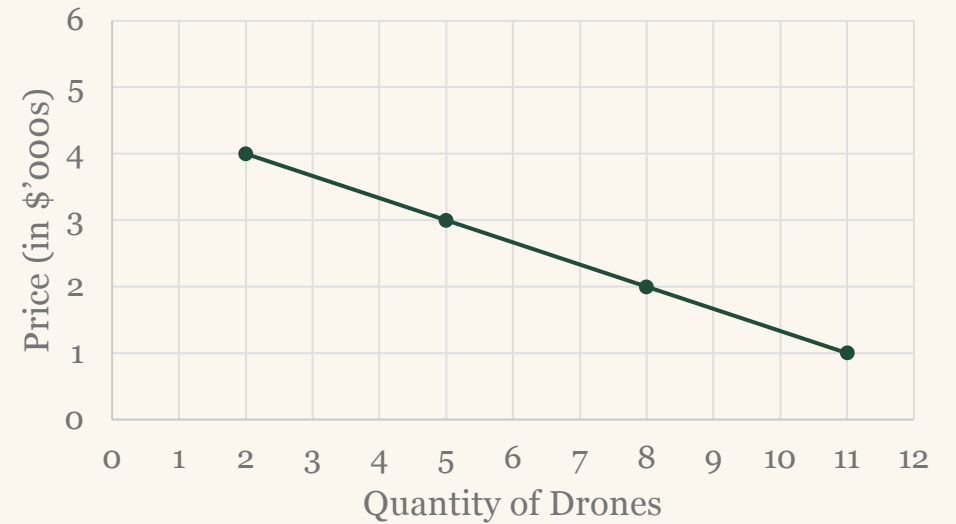
How large is the fall in quantity demanded?

When Buyers React Differently

Forensics

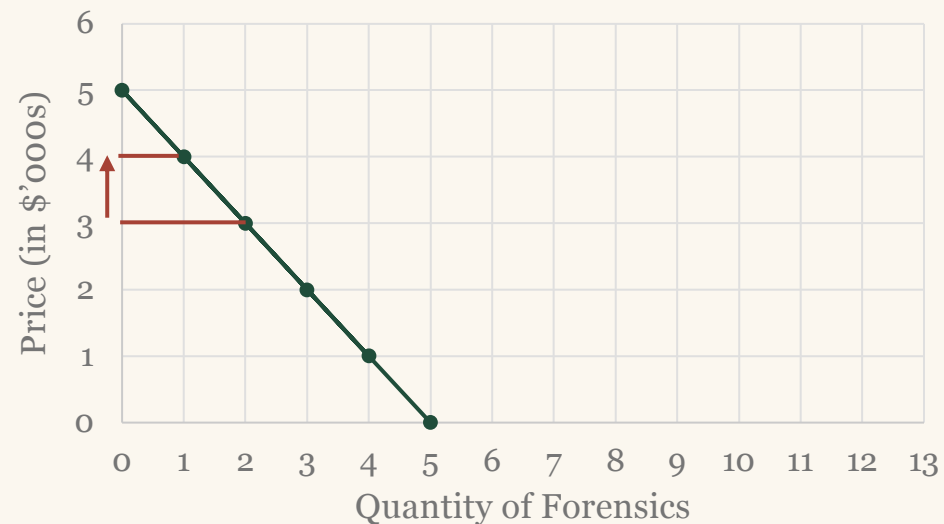


Drone Search Units



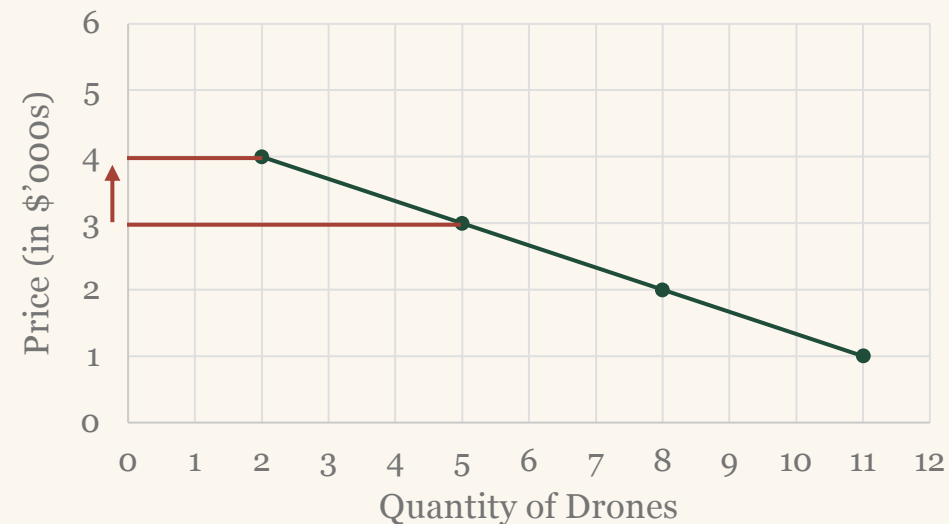
When Buyers React Differently

Forensics



Change in price from
3→4

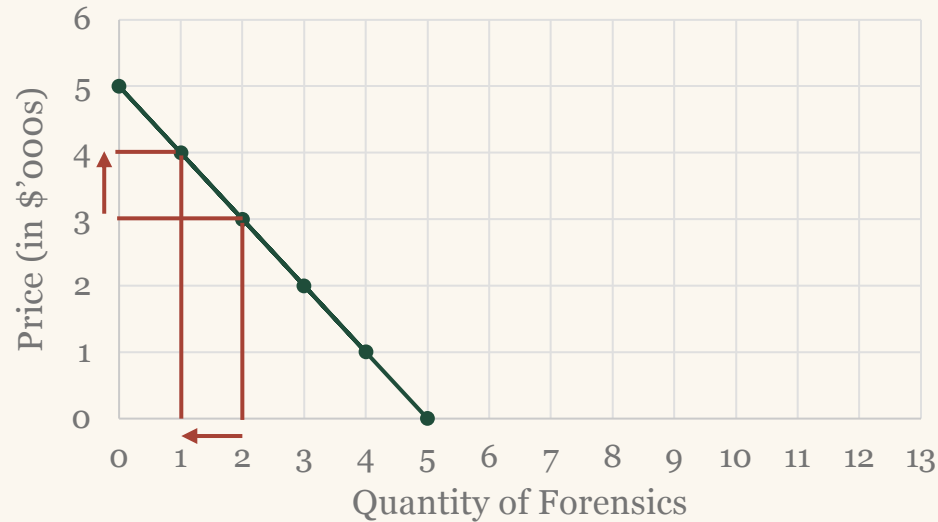
Drone Search Units



Change in price from
3→4

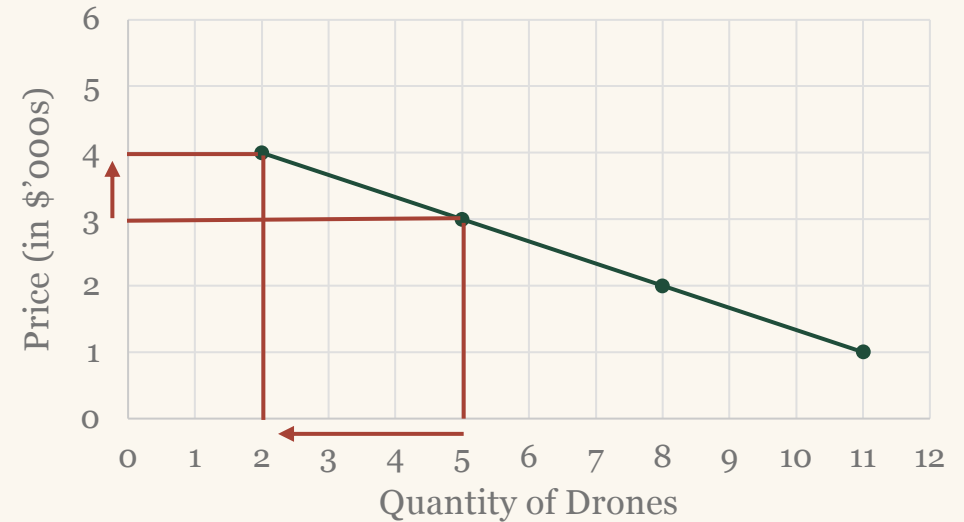
When Buyers React Differently

Forensics



Change in price from
3→4
Change in quantity demanded
2→1

Drone Search Units



Change in price from
3→4
Change in quantity demanded
5→2

Elasticity

Elasticity measures how much one economic variable responds to a change in another variable.

Price elasticity of demand measures how much quantity demanded responds when price changes.

Price Elasticity of Demand

$$\text{Price Elasticity of Demand} = \frac{\text{Percentage Change in Quantity Demanded}}{\text{Percentage Change in Price}}$$

Price and quantity change in opposite directions on the demand curve.

Price ↑ → Quantity Demanded ↓

Price Elasticity of Demand

$$\text{Price Elasticity of Demand} = \frac{\text{Percentage Change in Quantity Demanded}}{\text{Percentage Change in Price}}$$

Price and quantity change in opposite directions on the demand curve.

Price $\uparrow$ $\rightarrow$ Quantity Demanded $\downarrow$
Percentage Change in Price > 0

Price Elasticity of Demand

$$\text{Price Elasticity of Demand} = \frac{\text{Percentage Change in Quantity Demanded}}{\text{Percentage Change in Price}}$$

Price and quantity change in opposite directions on the demand curve.

Price $\uparrow$ $\rightarrow$ Quantity Demanded $\downarrow$

Percentage Change in Price $> 0 \rightarrow$ Percentage Change in Quantity Demanded < 0

Price Elasticity of Demand

$$\text{Price Elasticity of Demand} = \frac{\text{Percentage Change in Quantity Demanded}}{\text{Percentage Change in Price}}$$

Price and quantity change in opposite directions on the demand curve.

Price $\uparrow \rightarrow$ Quantity Demanded $\downarrow$

Percentage Change in Price $> 0 \rightarrow$ Percentage Change in Quantity Demanded < 0

Price Elasticity of Demand < 0

Mid-Point Formula

Price Elasticity of Demand (E_d) between points (P_1, Q_1) and (P_2, Q_2):

$$E_d = \underbrace{\left(\frac{Q_2 - Q_1}{\frac{Q_1 + Q_2}{2}} \right)}_{\substack{\text{Percentage} \\ \text{Change} \\ \text{in} \\ \text{Quantity} \\ \text{Demanded}}} \div \underbrace{\left(\frac{P_2 - P_1}{\frac{P_1 + P_2}{2}} \right)}_{\substack{\text{Percentage} \\ \text{Change} \\ \text{in} \\ \text{Price}}}$$

Mid-Point Formula

Price Elasticity of Demand (E_d) between points (P_1, Q_1) and (P_2, Q_2):

$$E_d = \left(\frac{Q_2 - Q_1}{\frac{Q_1 + Q_2}{2}} \right) \div \left(\frac{P_2 - P_1}{\frac{P_1 + P_2}{2}} \right)$$

- **Law of Demand:** If $P_2 > P_1$ then $Q_2 < Q_1$.
- Hence, $Q_2 - Q_1 < 0$ and $P_2 - P_1 > 0$.
- Thus, $E_d < 0$.

Mid-Point Formula

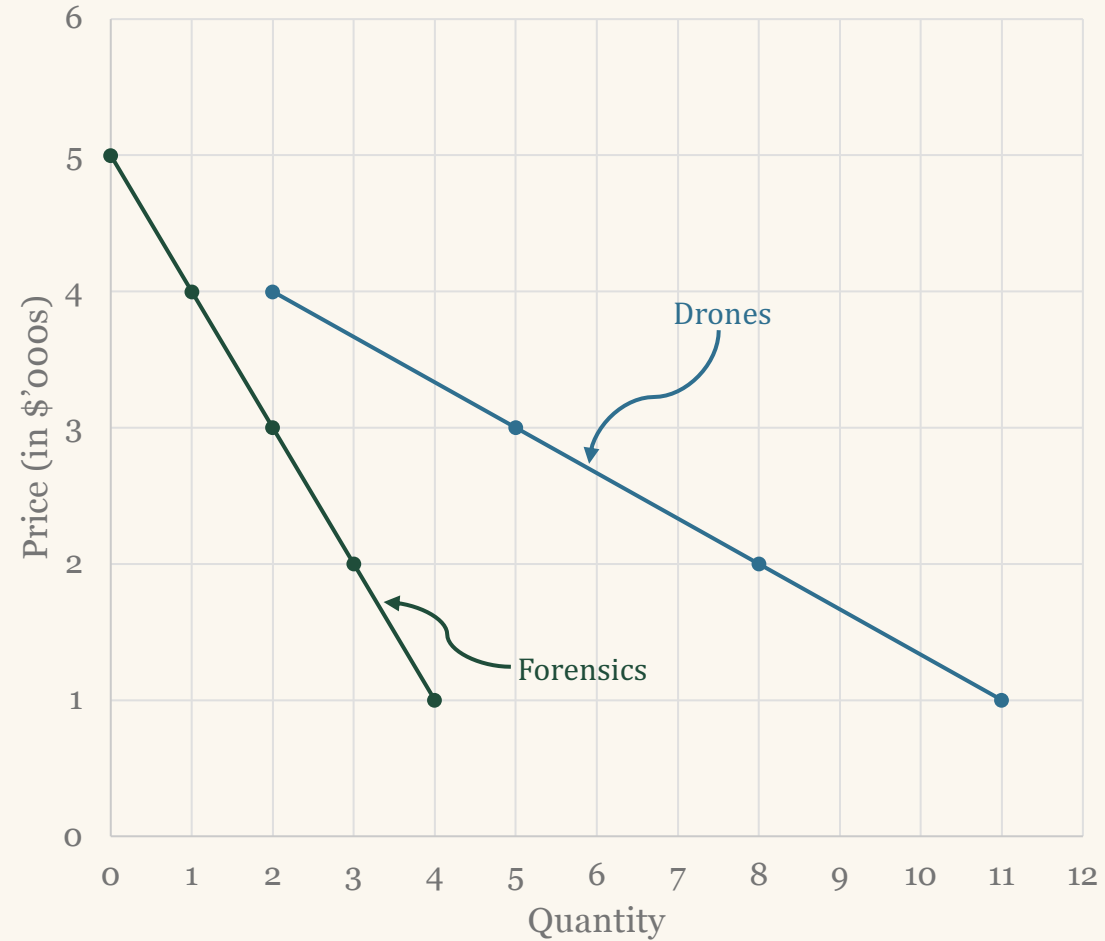
Price Elasticity of Demand (E_d) between points (P_1, Q_1) and (P_2, Q_2):

$$E_d = \left(\frac{Q_2 - Q_1}{\frac{Q_1 + Q_2}{2}} \right) \div \left(\frac{P_2 - P_1}{\frac{P_1 + P_2}{2}} \right)$$

- **Law of Demand:** If $P_2 > P_1$ then $Q_2 < Q_1$.
- Hence, $Q_2 - Q_1 < 0$ and $P_2 - P_1 > 0$.
- Thus, $E_d < 0$.

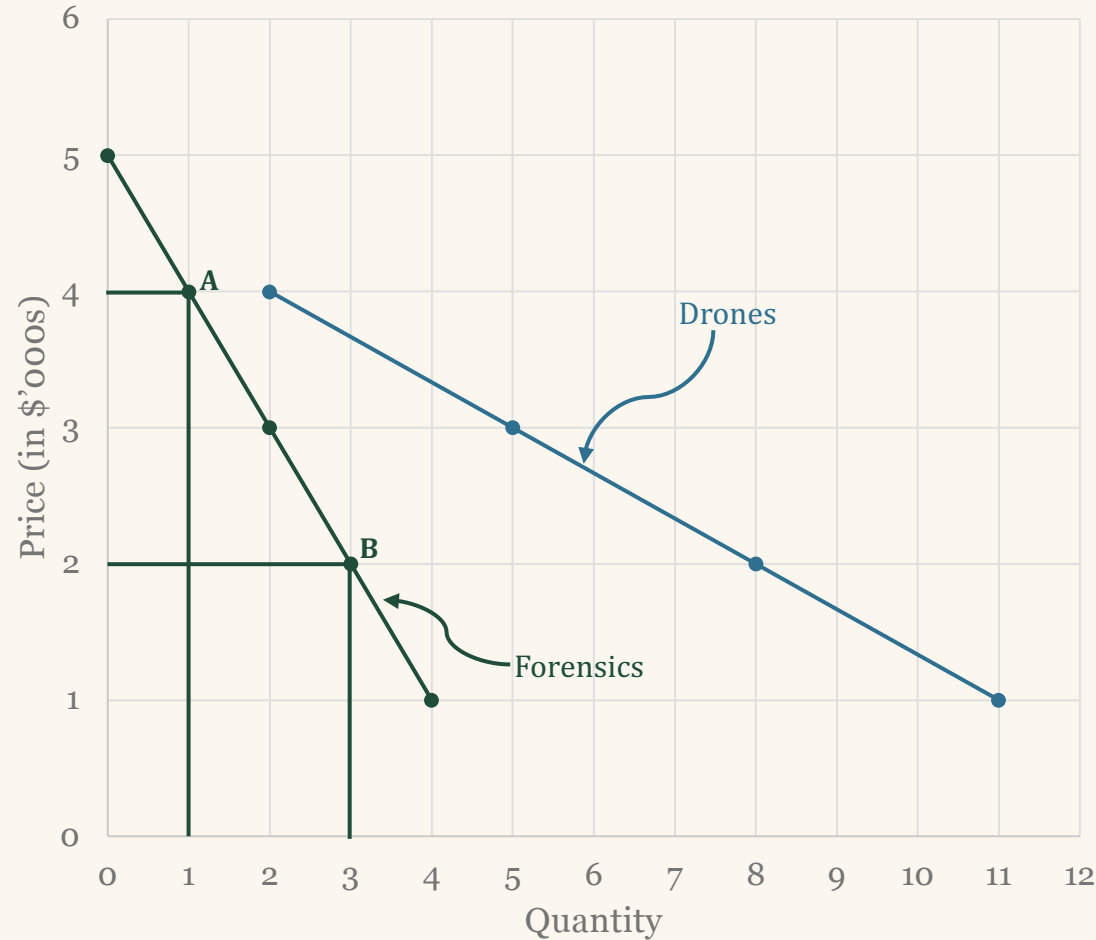
Sign of Price Elasticity of Demand is always negative.

Measuring the Sensitivity



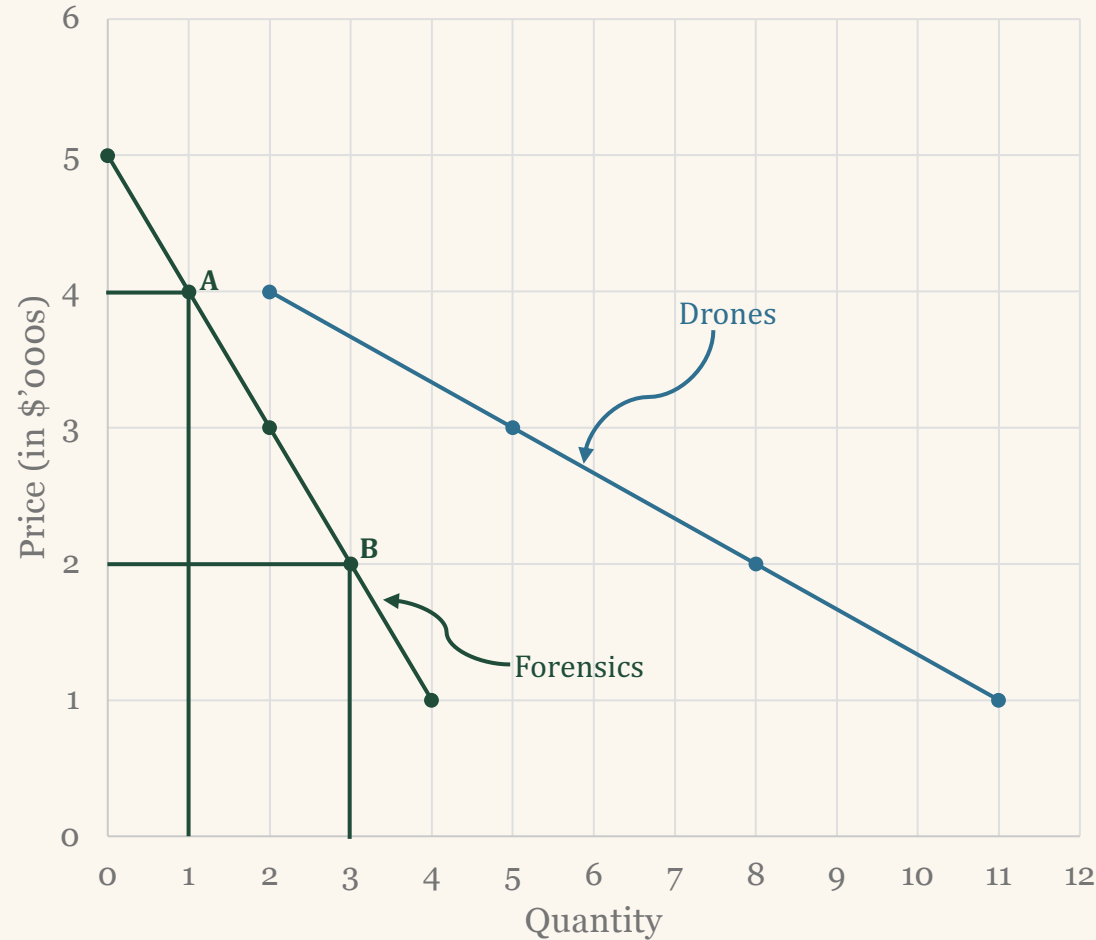
Measuring the Sensitivity

Price Elasticity of Demand between A and B in Forensics:



$$\begin{array}{ll} Q_1=1 & Q_2=3 \\ P_1=\$4000 & P_2=\$2000 \end{array}$$

Measuring the Sensitivity



Price Elasticity of Demand between A and B in Forensics:

$$Q_1=1 \quad Q_2=3$$
$$P_1=\$4000 \quad P_2=\$2000$$

Percentage Change in Quantity Demanded:

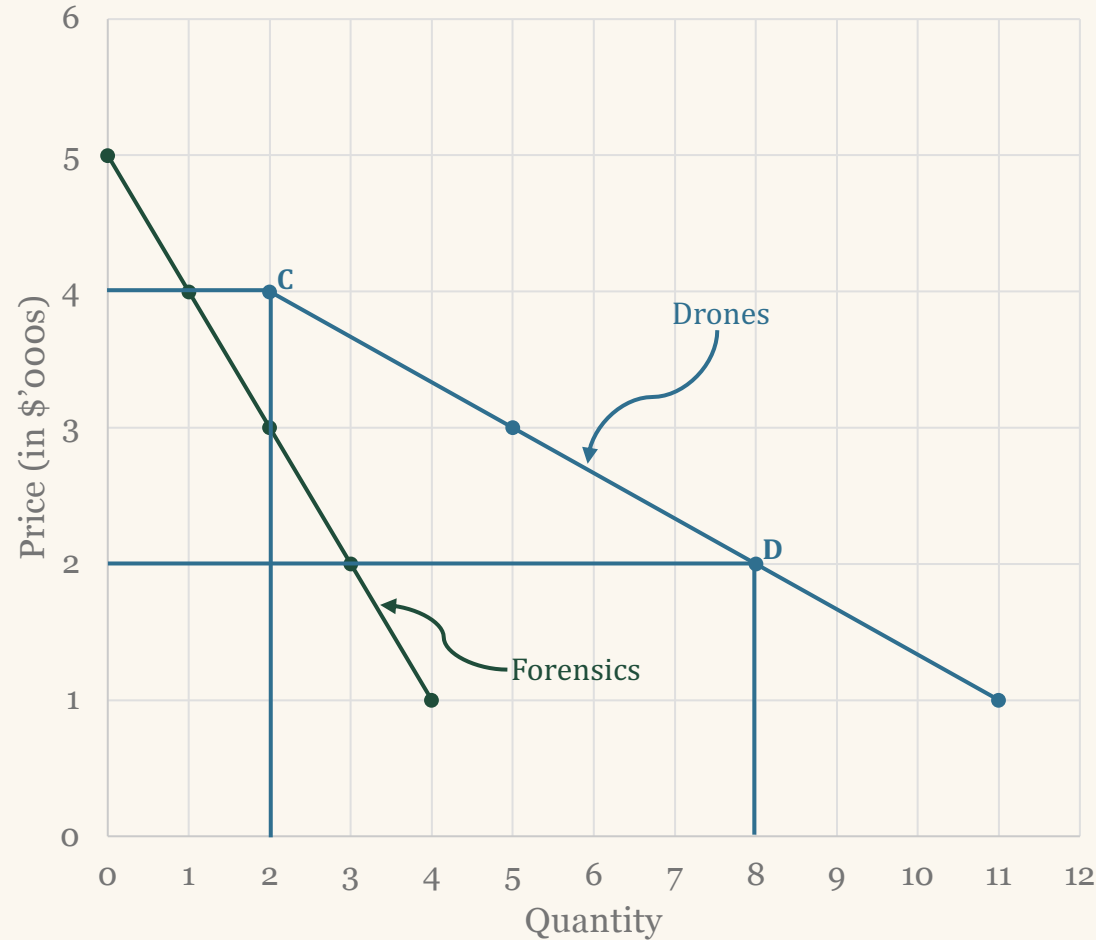
Percentage Change in Price:

Price Elasticity of Demand:

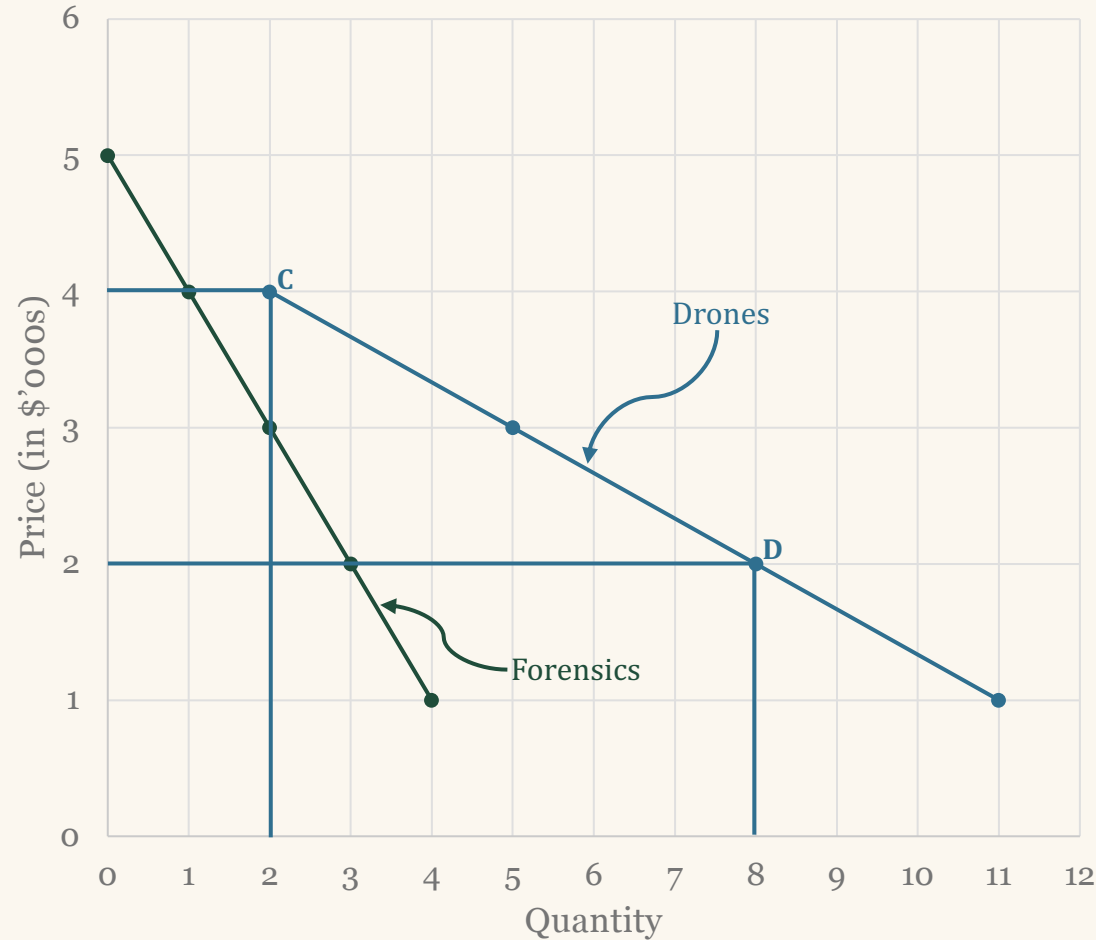
Measuring the Sensitivity

Price Elasticity of Demand between C and D for Drones:

$$\begin{array}{ll} Q_1=2 & Q_2=8 \\ P_1=\$4000 & P_2=\$2000 \end{array}$$



Measuring the Sensitivity



Price Elasticity of Demand between C and D for Drones:

$$Q_1=2 \quad Q_2=8$$
$$P_1=\$4000 \quad P_2=\$2000$$

Percentage Change in Quantity Demanded:

Percentage Change in Price:

Price Elasticity of Demand:

Elastic, Inelastic, Unit Elastic

- **Elastic:** quantity responds more than price.

$$|E_d| > 1$$

- **Inelastic:** quantity responds less than price.

$$|E_d| < 1$$

$$E_d = \frac{\% \Delta Q}{\% \Delta P}$$

- **Unit elastic:** quantity and price change by the same percentage.

$$|E_d| = 1$$

$$|\% \Delta Q| > |\% \Delta P|$$

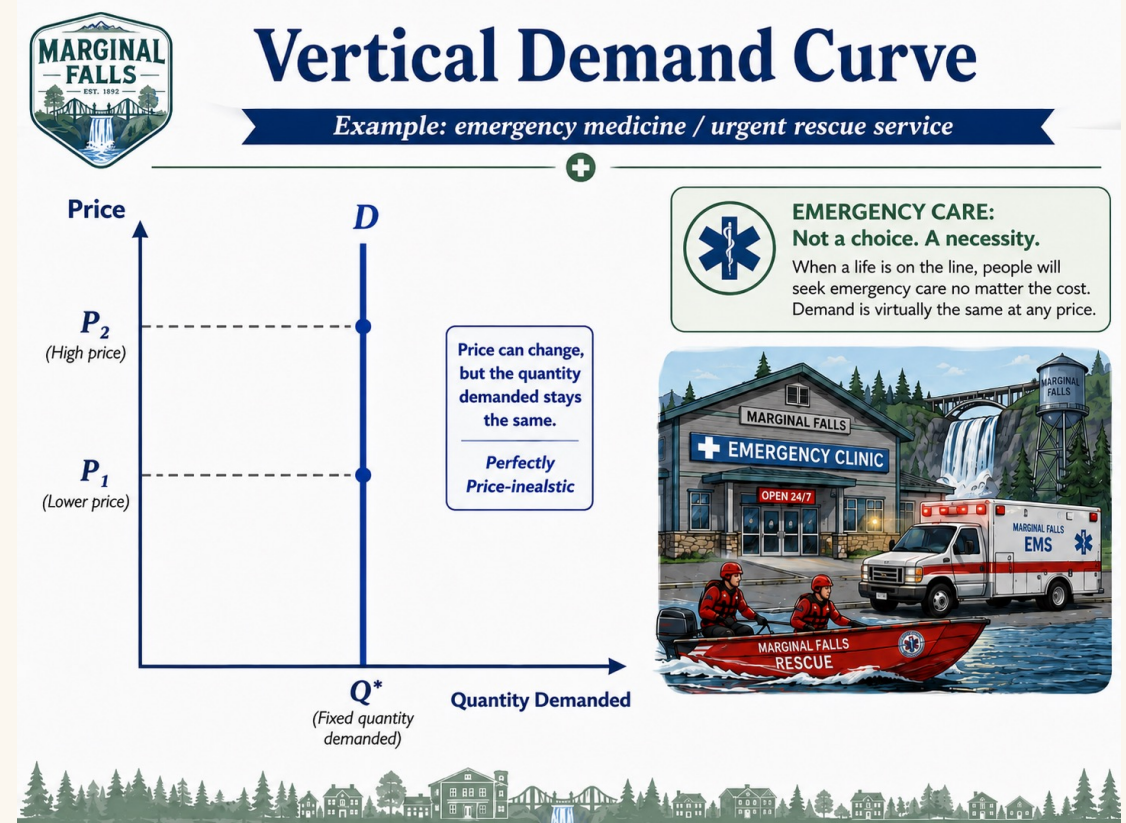
$$|\% \Delta Q| < |\% \Delta P|$$

$$|\% \Delta Q| = |\% \Delta P|$$

Perfectly Inelastic Demand

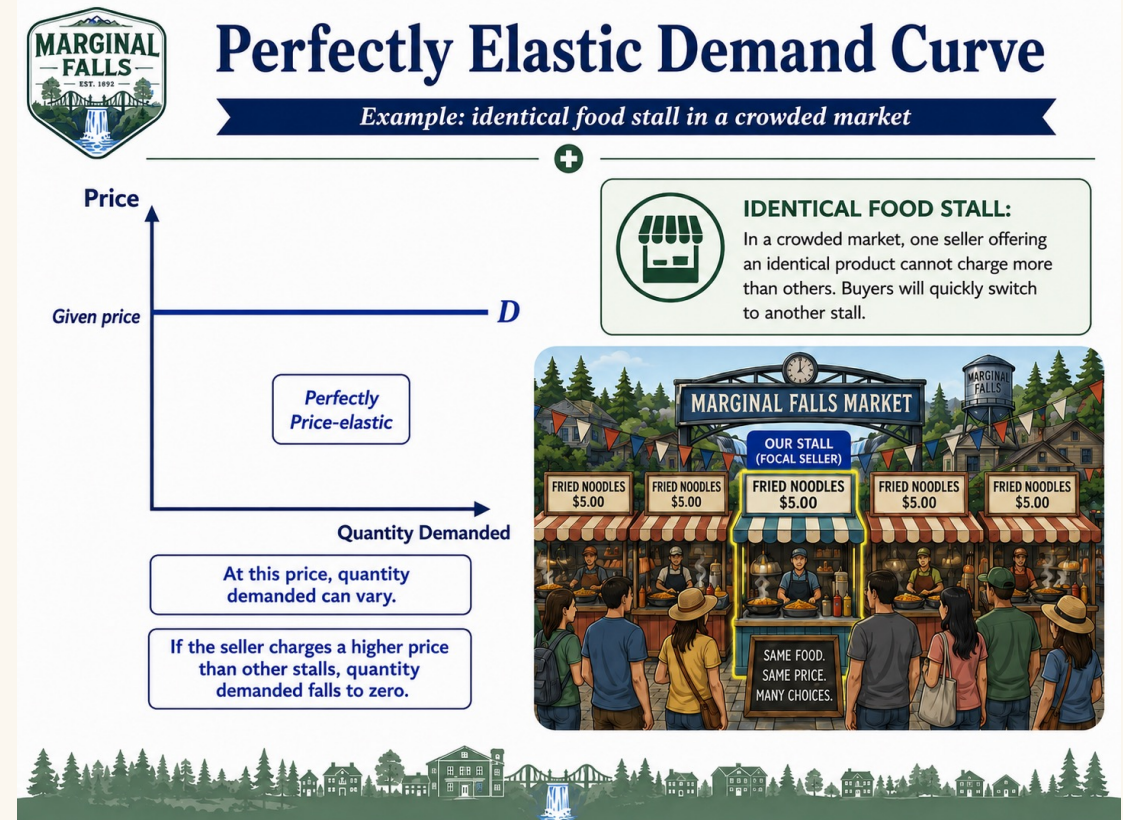
Perfectly inelastic demand means quantity demanded does not respond to price.

$$E_d = 0$$



Perfectly Elastic Demand

Perfectly elastic demand means buyers are infinitely responsive to price.



What Makes Demand More Elastic?

Substitutes

Demand is more elastic when buyers have close substitutes.

If bus fare rises, citizens can switch to the ferry and vice versa.

Easy to switch → more elastic.



What Makes Demand More Elastic?

Time

In the short run, buyers may be stuck.
In the long run, buyers can adjust.

Gas price rises!

Today: still take the car.

Later: buy electric car, move closer to work.

More time → more elastic demand.



What Makes Demand Less Elastic?

Luxury or Necessity

Demand is less elastic when buyers feel the good is necessary.

Daily bread: less elastic.

Festival cake: more elastic.

Hard to go without → less elastic.

Easy to postpone → more elastic.



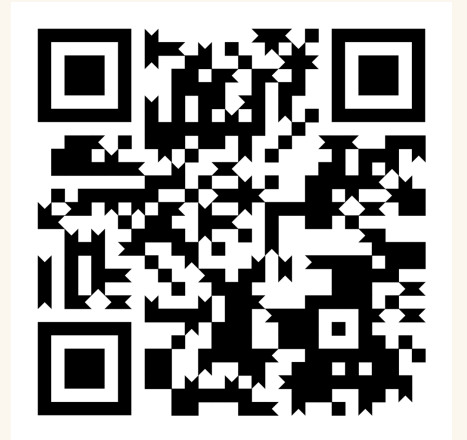
Junior Market Analyst

<https://econgames.promitkchaudhuri.com/elasticity>

You are analysts at the **Marginal Falls Market Desk**.

Sort each good into the correct jar:

Elastic or Inelastic



Cross-Price Elasticity: Measuring Relationships

Cross-price elasticity measures how quantity demanded of one good responds when the price of another good changes

Two goods A and B:

$$\text{Cross Price Elasticity of Demand for B} = \frac{\% \Delta Q \text{ for B}}{\% \Delta P \text{ for A}}$$

A and B are substitutes → Cross Price Elasticity of Demand for B > 0

A and B are complements → Cross Price Elasticity of Demand for B < 0

End of Act II